

Fetch a Joke — Using Hooks & Connections

Code:

```
from airflow import DAG
from airflow.providers.standard.operators.python import PythonOperator
from airflow.providers.http.hooks.http import HttpHook
from datetime import datetime

def get_random_joke(**kwargs):
    hook = HttpHook(method="GET", http_conn_id="joke_api")
    response = hook.run("/random_joke")

    joke = response.json()

    print(f"Type: {joke['type']}")
    print(f"Setup: {joke['setup']}")
    print(f"Punchline: {joke['punchline']}")

    return joke["setup"]

def get_ten_jokes(**kwargs):
    hook = HttpHook(method="GET", http_conn_id="joke_api")
    response = hook.run("/random_ten")

    jokes = response.json()
    count = len(jokes)

    print(f"Fetches {count} jokes")

    if count > 0:
        print(f"First setup: {jokes[0]['setup']}")

    return count

def show_results(**kwargs):
    ti = kwargs['ti']

    setup = ti.xcom_pull(task_ids="get_random_joke")
    count = ti.xcom_pull(task_ids="get_ten_jokes")

    print(f"Random joke setup: {setup}")
    print(f"Total jokes fetched across both tasks: {1 + count}")
    print("Connection used: joke_api – no URL written in this file.")

with DAG(
    dag_id="joke_fetch_demo",
```

```

    schedule="@once",
    start_date=datetime(2024, 1, 1),
    catchup=False
) as dag:

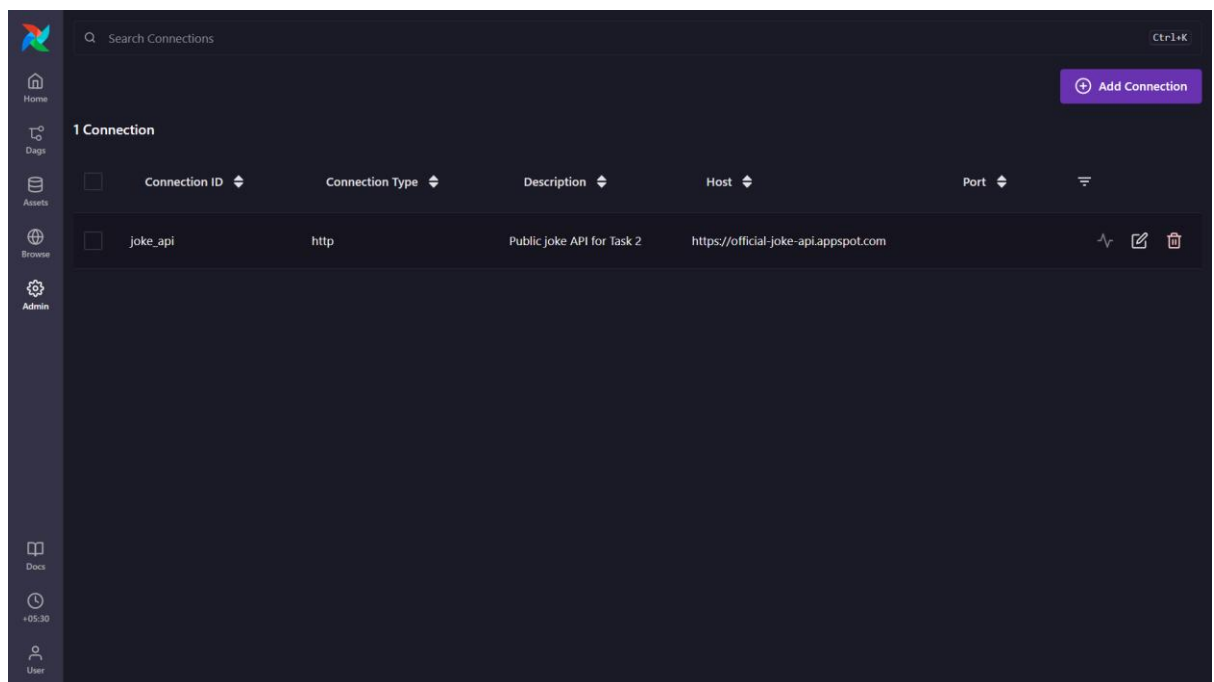
    get_random_joke = PythonOperator(
        task_id="get_random_joke",
        python_callable=get_random_joke
    )

    get_ten_jokes = PythonOperator(
        task_id="get_ten_jokes",
        python_callable=get_ten_jokes
    )

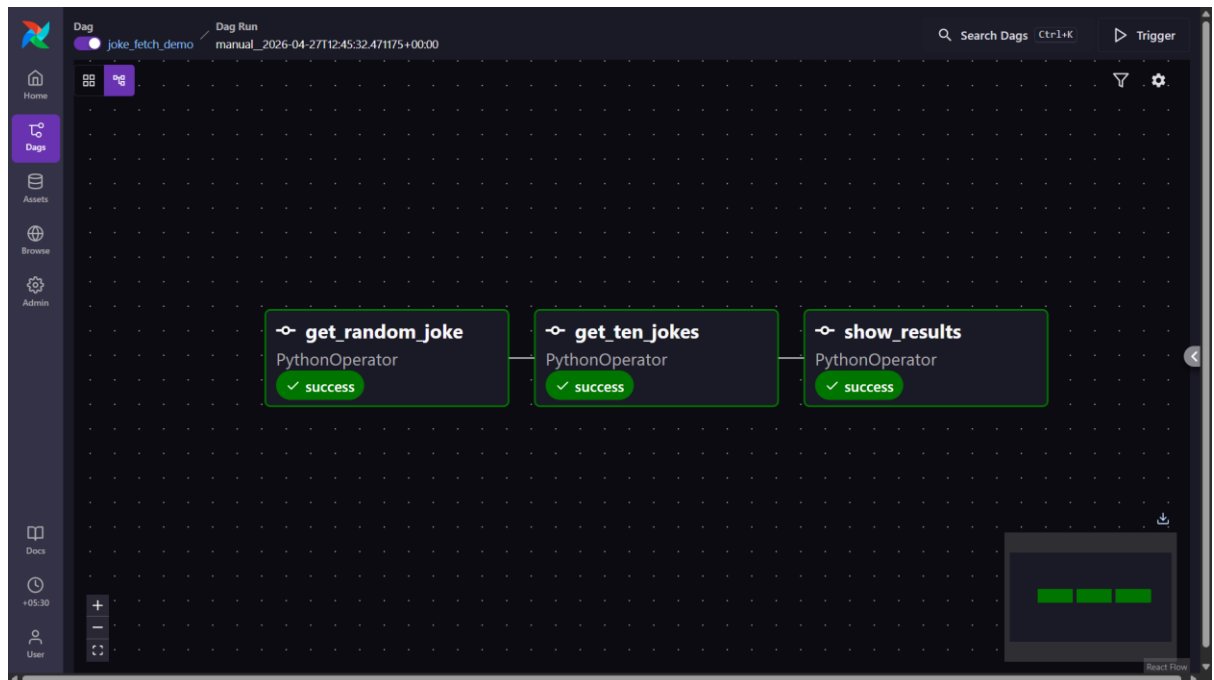
    show_results = PythonOperator(
        task_id="show_results",
        python_callable=show_results
    )
    get_random_joke >> get_ten_jokes >> show_results

```

1. Admin → Connections page — showing joke_api listed with Connection Type = HTTP



2. Graph View — all 3 tasks GREEN: get_random_joke → get_ten_jokes → show_results



3. Task Log of get_random_joke — showing the joke type, setup, and punchline printed

The image shows the Airflow Task Log for the 'get_random_joke' task. The task is in a 'Success' state. The log messages are displayed in a table with columns for 'Log Level', 'Message', and 'Source'. The log messages show the task's execution details, including the joke type, setup, and punchline. The log messages are as follows:

Log Level	Message	Source
INFO	Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/joke_fetch.py	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	TaskInstance Details: dag_id=joke_fetch_demo task_id=get_random_joke dagrun_id=manual_2026-04-27T12:45:32.471175+00:00	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	Type: general	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	Setup: Do you know where you can get chicken broth in bulk?	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	Punchline: The stock market.	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	Done. Returned value was: Do you know where you can get chicken broth in bulk?	/usr/local/airflow/dags/Assignment-2/joke_fetch.py
INFO	Pushing xcom ti=RuntimeTaskInstance(id=UUID('019dcf02-0bcf-78ba-9f10-9b4cb7bc8405'), task_id=get_random_joke, dag_id=joke_fetch_demo, run_id=manual_2026-04-27T12:45:32.471175+00:00)	/usr/local/airflow/dags/Assignment-2/joke_fetch.py

4. Task Log of show_results — showing the combined totals and the "no URL" confirmation line

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, Admin, Docs, and a clock showing 05:30. The main area shows a DAG named 'joke_fetch_demo' with a 'Dag Run' status of 'manual_2026-04-27T12:55:11.040485+00:00'. The task 'show_results' is highlighted, showing a 'Success' status. Below the task name, a table lists the operator as 'PythonOperator', the start date as '2026-04-27 18:25:13', the end date as '2026-04-27 18:25:13', the duration as '00:00:00.243', and the DAG version as 'v3'. The 'Logs' tab is selected, showing a list of log messages. A red box highlights the following log entries:

```
2 [2026-04-27 18:25:13] WARNING - Astro managed secrets backend is disabled
3 [2026-04-27 18:25:13] INFO - Data bundles loaded: dags-folder
4 [2026-04-27 18:25:13] INFO - Filling up the DagBag from /usr/local/airflow/dags/Assignment-2/joke_fetch.py
5 [2026-04-27 18:25:13] INFO - TaskInstance Details: dag_id=joke_fetch_demo task_id=show_results dagrun_id=manual_2026-04-27T12:55:11.040485+00:00
6 [2026-04-27 18:25:13] INFO - Random joke setup: Do you know where you can get chicken broth in bulk?
7 [2026-04-27 18:25:13] INFO - Total jokes fetched across both tasks: 11
8 [2026-04-27 18:25:13] INFO - Connection used: joke_api - no URL written in this file.
9 [2026-04-27 18:25:13] INFO - Done. Returned value was: None
```